

Modelling effects of root growth and decay on soil water retention and permeability

J.J. Ni, A.K. Leung, and C.W.W. Ng

Abstract: Plant roots can change the soil water retention curve (SWRC) and saturated permeability (k_{sat}) of vegetated soils. However, there is no model that could capture both the effects of root growth and root decay on these soil hydraulic properties simultaneously. This note proposes a new void ratio function that can model the decrease and increase in soil void ratio due to root occupancy (upon growth) and root shrinkage (upon decay), respectively, in an unsaturated vegetated coarse-grained soil. The function requires two root parameters; namely, root volume ratio and root decay ratio, both of which can be readily measured through root excavation and image-based analysis. The new function is incorporated into a void ratio-dependent SWRC model for predicting the SWRC of vegetated soils. Similarly, the same function can be combined with the Kozeny–Carman equation for predicting k_{sat} . The model prediction is then compared with a set of new field test data and an existing laboratory dataset for a silty sand vegetated with plant species under the family *Schefflera*. Good agreements are obtained between the measurements and predictions.

Key words: partial saturation, permeability, suction, vegetation, water flow.

Résumé : Les racines des plantes peuvent modifier la courbe de rétention d'eau du sol (« SWRC ») et la perméabilité saturée (k_{sat}) des sols végétalisés. Cependant, aucun modèle ne permet de prendre en compte simultanément les effets de la croissance et de la pourriture racinaire sur les propriétés hydrauliques du sol. Cette note propose une nouvelle fonction de l'indice des vides qui peut modéliser la diminution et l'augmentation de l'indice des vides du sol en raison de l'occupation des racines (lors de la croissance) et du rétrécissement des racines (après la pourriture), respectivement, dans un sol à grain grossier végété non saturé. La fonction nécessite deux paramètres de racine, à savoir le rapport volume racine et le rapport de décomposition de la racine, qui peuvent tous deux être facilement mesurés par le biais d'une excavation des racines et d'une analyse par image. La nouvelle fonction est intégrée dans un modèle SWRC en fonction de l'indice des vides pour prédire le SWRC des sols végétalisés. De même, la même fonction peut être combinée avec l'équation de Kozeny–Carman pour prédire le k_{sat} . La prévision du modèle est ensuite comparée à un ensemble de nouvelles données d'essai sur le terrain et à un jeu de données de laboratoire existant pour un sable limoneux couvert de végétation avec des espèces végétales de la famille *Schefflera*. De bons accords sont obtenus entre les mesures et les prévisions. [Traduit par la Rédaction]

Mots-clés : saturation partielle, perméabilité, succion, végétation, écoulement de l'eau.

Introduction

Recent studies have evidently demonstrated that the presence of plant roots could affect both the soil water retention curve (SWRC) and permeability. Comparisons of SWRCs of bare and vegetated soils show that the soil water retention capability could be increased or decreased, depending on plant type (Scholl et al. 2014; Rahardjo et al. 2014; Leung et al. 2015a; Yan and Zhang 2015), plant spacing (Ng et al. 2016b), and plant age (Jotisankasa and Sirirattanachai 2017). Plant roots would also reduce (Gish and Jury 1983; Leung et al. 2015a; Jotisankasa and Sirirattanachai 2017) or increase (van Noordwijk et al. 1991; Mitchell et al. 1995; Vergani and Graf 2016; Ni et al. 2017; Leung et al. 2018) saturated permeability (k_{sat}) or infiltration rate. The increase in soil water retention and the decrease in k_{sat} were believed to be associated with the reduction of soil void ratio upon root growth and occupancy of soil pore space (Scholl et al. 2014; Leung et al. 2015b; Ng et al.

2016b). On the contrary, formation of soil macropores (i.e., increase in void ratio) due to root shrinkage upon decay (Jassogne et al. 2007; Ghestem et al. 2011) was thought to be related to the decrease in soil water retention and the increase in k_{sat} . Indeed, although thick structural roots may persist for many years, finer roots (diameter < 2 mm) may be born and die within a period of weeks or months (Eissenstat et al. 2000; Block et al. 2006).

In general, root-induced changes in soil hydraulic properties were rarely considered in most existing transient seepage analysis for geotechnical infrastructure. This is primarily due to the lack of capability of proper modelling the soil–water–root interaction. Existing models (e.g., Scanlan and Hinz 2010; Scholl et al. 2014) contain a number of empirical parameters that are not easy to obtain and calibrate. Ng et al. (2016a) proposed a SWRC model for vegetated soils, but the model only considers the effects of root growth, not root decay.

Received 18 June 2018. Accepted 3 October 2018.

J.J. Ni. Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Hong Kong SAR.

A.K. Leung.* Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Hong Kong S.A.R.; formerly Division of Civil Engineering, School of Science and Engineering, University of Dundee, Dundee, Scotland, UK.

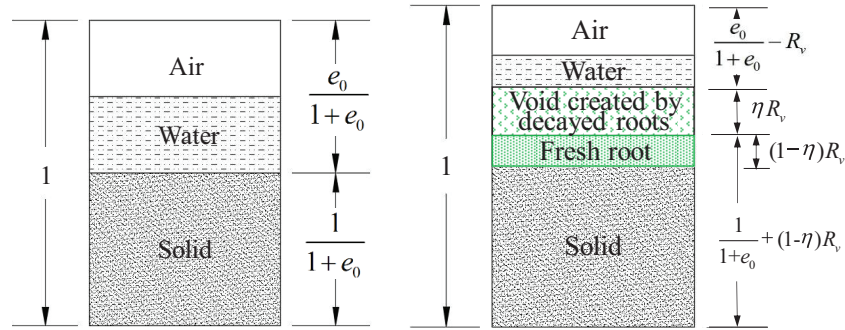
C.W.W. Ng.* The Hong Kong University of Science and Technology, Hong Kong S.A.R.

Corresponding author: A.K. Leung (email: ceanthony@ust.hk).

*A.K. Leung currently serves as an Editorial Board Member; C.W.W. Ng currently serves as an Associate Editor; peer review and editorial decisions regarding this manuscript were handled by G. Siemens.

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Fig. 1. Phase diagram of unsaturated vegetated soil considering root growth and root decay (modified from Ng et al. 2016a). [Colour online.]



This note proposes and validates some new models that could capture the effects of both root occupancy and root decay on SWRC and k_{sat} of unsaturated vegetated soils.

Model formulation

Before root decay, an unsaturated vegetated soil contains not only air, water, and solid phases, but is considered to have another phase associated with root occupancy in soil pore space (Ng et al. 2016a; Fig. 1). The void ratio of the vegetated soil may be expressed as

$$(1) \quad e = \frac{e_0 - (1 - \eta)R_v(1 + e_0)}{1 + (1 - \eta)R_v(1 + e_0)}$$

where e_0 is the void ratio of bare soil; η is the root decay ratio or fraction of the “return” of the soil pore space (in volume) due to root shrinkage upon decay, or known as root decay ratio in this study; and R_v is the root volume ratio (mm^3/mm^3), which is defined as the total volume of roots per unit volume of soil. Upon decay, roots would shrink when they lose the ability to uptake soil moisture (Ghestem et al. 2011; Brunner et al. 2015). A portion of root biomass would be decomposed and then turn into organic matter (Jacobs et al. 2018). These biochemical processes made the decayed roots to appear hollow (Ni et al. 2017). Indeed, when compared to fresh roots of shrub and grass species investigated in this study, the dry biomass of decayed roots reduced by more than 80% (shown later). Smith et al. (2014) also observed that upon root decay, the root biomass of the majority of the graminoid species had reduced by 77% within the first few months of plant death. Hence, it is a simplified, yet reasonable, approach to model the effects of root decay on (i) root volumetric shrinkage and (ii) loss of dry root biomass (upon biochemical decomposition) by considering an increase in soil pore space using the fraction η , as shown in Fig. 1. $R_v = 0$ means that there are no roots in the soil (i.e., bare soil). R_v cannot be larger than $e_0/(1+e_0)$, which means all soil pore space has been fully occupied by roots. Plant roots can shrink as root water potential or root suction increases. Roots started to shrink only when root suction exceeded 1 MPa (Boldrin et al. 2017, 2018). As the range of soil suction investigated in this study was less than 100 kPa (shown later), any root volume change due to soil suction can be reasonably neglected. When there is no root decay ($\eta = 0$), eq. (1) resembles the void ratio equation proposed by Ng et al. (2016a). Strictly speaking, η should not be equal to 1, as certain volume of root biomass would remain in the pore space even if the roots are fully decayed. Previous experiments show that a plant root could reduce its diameter by up to 30% from fully hydrated to fully dry state (Boldrin et al. 2017, 2018). There should exist an upper bound η , but quantifying this upper bound value requires further understanding of root biomechanics of a specific plant species.

To model the effects of roots on the change in water retention ability of a soil, the void ratio-dependent SWRC equation proposed by Gallipoli et al. (2003) is adopted

$$(2) \quad S_r = \left[1 + \left(\frac{se^{m_4}}{m_3} \right)^{m_2} \right]^{-m_1}$$

where S_r is the soil degree of saturation; s is the matric suction; and m_1 , m_2 , m_3 (kPa), m_4 are the model parameters. m_1 and m_2 control the shape of a SWRC (van Genuchten 1980), while m_3 and m_4 are related to the air-entry value (AEV) of the bare soil. Considering that void ratio of coarse-grained soils has negligible effects on SWRC at high suction range, the product, $m_1 m_2 m_4$, is fixed to be 1 (Gallipoli et al. 2003). To capture the reduction of AEV due to the formation of soil macropores upon root decay (Ghestem et al. 2011; Vergani and Graf 2016; Ng et al. 2016b; Ni et al. 2017), eq. (2) may be modified as

$$(3) \quad S_r = \left\{ 1 + \left[\frac{se^{m_4}}{m_3 \exp\left(-k \frac{e_r}{e_{r0}}\right)} \right]^{m_2} \right\}^{-m_1}$$

where k is a model parameter that controls the rate of reduction of the AEV due to root decay and this parameter is dependent on the plant type and soil type; e_r is the macro void ratio “returned” from root decay

$$(4) \quad e_r = \frac{\eta R_v(1 + e_0)}{1 + (1 - \eta)R_v(1 + e_0)}$$

and e_{r0} is the void ratio occupied by roots (i.e., $e_0 - e$), when $\eta = 0$.

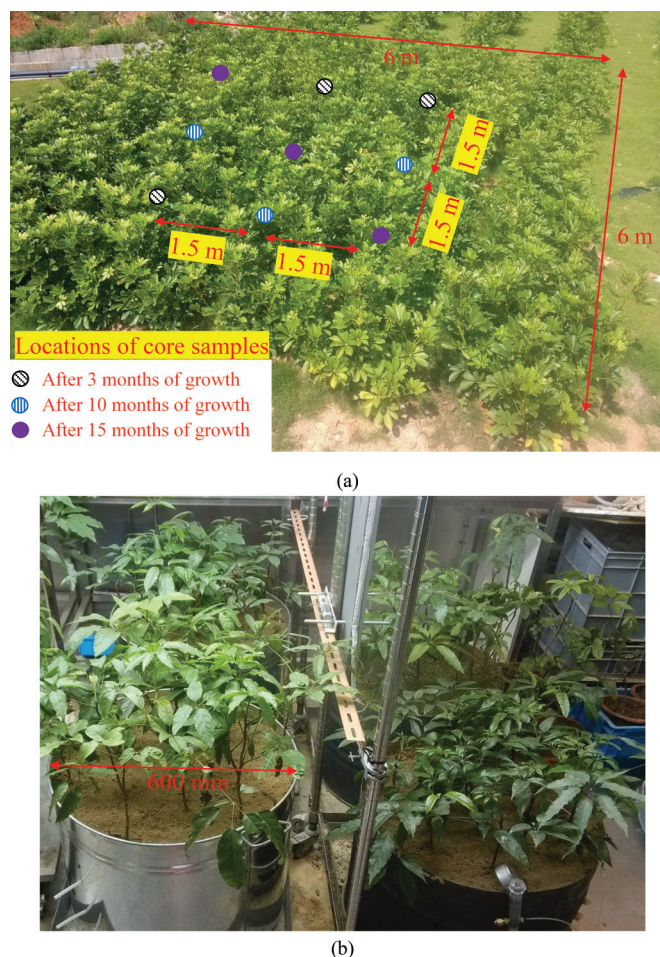
By comparing eq. (3) with that proposed by Ng et al. (2016a), two additional parameters η and k are needed to take into account both the effects of root occupancy and (or) root decay on SWRC simultaneously. When $\eta = 0$, eq. (1) resembles that proposed by Ng et al. (2016a). Hence, by knowing the SWRC of the bare soil, the root volume ratio (R_v), the decay ratio η , and the parameter k , the SWRC of an unsaturated vegetated soil can be predicted.

To model the effects of the root on k_{sat} , the Kozeny–Carman (KC) equation was used to estimate the k_{sat} of coarse-grained soil (Carman 1938). The KC equation can be expressed as (Carrier 2003)

$$(5) \quad k_{\text{sat}} = A \frac{e^3}{1 + e}$$

where A is a factor that depends on pore shape and tortuosity of water flow channels, wetted surface per unit volume of soil particles, the unit weight of water, and the viscosity (Carrier 2003). To

Fig. 2. (a) Overview of test setup and locations of core samples in field test; (b) overview of test setup for the laboratory tests conducted by Ng et al. (2016b). [Colour online.]



capture the effects of plants, the parameter A in eq. (5) may be modified as follows:

$$(6) \quad A' = \frac{A}{\exp\left(-k \frac{e_r}{e_{r0}}\right)}$$

When $\eta = 0$, A' resembles A . A can be calibrated based on known e and k_{sat} of a bare soil. Hence, by knowing R_v and η , k_{sat} of vegetated soils at any decay ratio can be determined. Although the KC equation was originally developed for coarse-grained soils, subsequent studies have shown that the equation may also be applied to other types of soils (e.g., clay), which has a range of k_{sat} between 10^{-12} and 10^{-2} m/s (Chapuis and Aubertin 2003; Urumović and Urumović 2016). Indeed, when calibrating the coefficient, A , in eq. (5), the effects of fine particles on the tortuosity of water flow channels and specific surface area per unit volume of soil particle have already been taken into account empirically.

Model validation

Test plan and procedures

To validate the new models, a new field test was conducted and an existing laboratory study reported by Ng et al. (2016b) was used. The field test was performed at a site at Shenzhen, China (Fig. 2a) on a 6 m x 6 m flat plot between 2016 and 2017. The field soil tested was compacted completely decomposed granite (CDG), which has

18% gravel, 52.7% sand, 18.6% silt, and 10.7% clay. CDG is a soil type typically found in tropical and subtropical regions of the world, such as Hong Kong, South Korea, and Brazil. This soil type is therefore commonly used as a construction material for man-made slopes, where native vegetation would be planted (GEO 2011). The liquid limit and plasticity index of the CDG are 33% and 18%, respectively. According to the Unified Soil Classification System (USCS; ASTM 2010a), the field soil can be classified as clayey sand (SC). The CDG was compacted to a dry density of 1670 kg/m^3 , which corresponds to 95% of the maximum dry density. After compaction, six soil cores were sampled in random locations in the plot to determine in situ dry density and k_{sat} , following ASTM (2010b) standard D5084. A cylindrical core cutter (76 mm in diameter and 100 mm in height) was penetrated vertically into the ground to a depth of 100 mm for sampling. Both the dry density and k_{sat} varied slightly from 1638 to 1670 kg/m^3 (less than 2%) and from 5.7×10^{-5} to $6.3 \times 10^{-5} \text{ m/s}$ (less than 10%), respectively. These measurements suggest that the spatial variability was minimal in the field. Any subsequent change in the k_{sat} after transplantation would well be due to the effects of root growth or decay. The contents of nutrient, nitrogen (N) and phosphorus (P), of the field soil were 66.21 ± 1.98 and $36.26 \pm 0.69 \text{ mg/kg}$, respectively (Chen et al. 2018). After 3, 10, 15 months of plant growth, the content of N in the vegetated soil decreased to 62.3 ± 1.64 , 56.25 ± 1.78 , and $51.34 \pm 2.03 \text{ mg/kg}$, respectively, while that of P reduced to 34.42 ± 1.78 , 31.26 ± 1.92 , and $29.98 \pm 2.16 \text{ mg/kg}$, respectively.

The field plot was fully covered with grass turf (*Cynodon dactylon*) in early February 2016. Grass turf with a shoot length of $50 \pm 9 \text{ mm}$ and a root depth of $20 \pm 4 \text{ mm}$ was obtained from a nursery. Then, *Schefflera arboricola* (a shrub) was transplanted to the plot on 2 April 2016. All *S. arboricola* transplanted has a root depth of $200 \pm 24 \text{ mm}$. It has been shown by previous studies that the shrub and grass species chosen could survive in coarse-grained media (e.g., CDG) both in the laboratory and in the field (Leung et al. 2015a; Ng et al. 2016b; Ni et al. 2017).

Three pairs of moisture probe (SM300 from Delta-T Device Ltd.) and jet-fill tensiometers (JFTs) were installed at 100 mm depth for monitoring the responses of soil volumetric water content (VWC) and matric suction, respectively. Prior to installation, the ceramic tip and plastic tube of each JFT were fully saturated and completely filled with de-aired water. Distilled water was used when refilling all the JFTs. A small concentration of algicide with blue color was added to the distilled water to prevent the growth of algae in each tensiometer (Soilmoisture Equipment Corporation 1997). Measurements by the soil moisture probes are based on the principle of time domain reflectometry (Whalley 1993). Each probe generates an electromagnetic pulse to polarize the surrounding soil. The time required for the polarization depends on the soil apparent dielectric constant, which is governed by the soil VWC. More detailed working principle can be found in Delta-T Devices Ltd. (2010). Disturbed soil samples were collected from the field after 0, 3, 10, and 15 months of plant transplantation. Based on the loss-on-ignition method (Ball 1964), the soil organic content is found to be 1.2%, 2.6%, 2.8%, and 3.3%, respectively. Each soil moisture probe was subsequently calibrated against any effects of organic content. The soil samples were compacted in multiple acrylic cylinders to the same average in situ dry density, but at different moisture content. Each soil moisture sensor was inserted into the soil and a steady state output (in voltage) was recorded. The voltage was then correlated with the corresponding VWC. A linear relation is found (see Fig. A1 in Appendix A). The soil organic content does not affect the slope of the relation, but seems to introduce a shift of the relation, though deemed negligible by considering the sensor accuracy of 3% of VWC (Ng et al. 2016b). An average calibration curve was thus applied to all sensors for the entire monitoring period.

The field plot was left exposed to the natural environment for nearly 16 months from 2 April 2016 to 17 July 2017, during which

Fig. 3. Decayed and fresh roots excavated from the field test plots. [Colour online.]

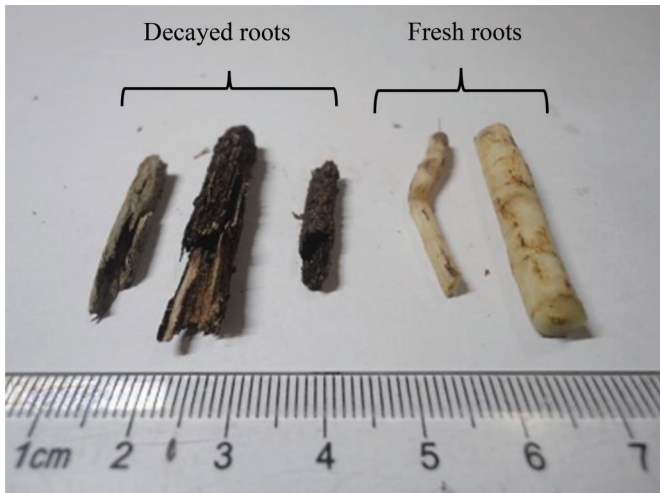
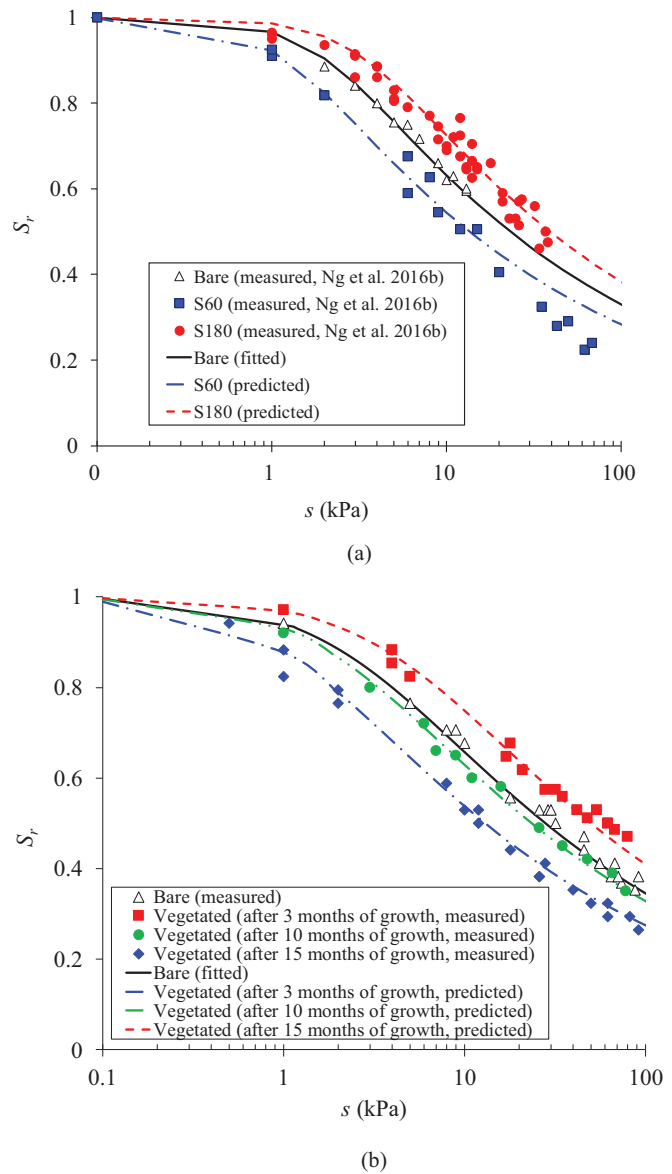


Table 1. Measured root volume ratio (R_v) and root decay ratio (η), together with their statistical test results.

Test condition	R_v (mm ³ /mm ³)	η (%)
Laboratory		
Spacing 60 mm	0.044 ± 0.004	11.0 ± 1.2
Spacing 180 mm	0.034 ± 0.005	1.0 ± 1.0
p-value	0.054	<0.001
Field		
After 3 months of growth	0.040 ± 0.006	1.2 ± 1.0
After 10 months of growth	0.053 ± 0.010	8.1 ± 1.2
After 15 months of growth	0.069 ± 0.008	16.0 ± 1.9
p-value	0.070	<0.001

soil matric suction and VWC were measured continuously. After 3, 10, and 15 months of growth, drying SWRCs of vegetated soils were determined by relating the VWC and suction during no-rain periods. S_r was calculated by dividing measured VWC over soil porosity. At the end of each growth period, three soil cores (76 mm in diameter) between 0 and 100 mm depths were sampled from the plot to measure k_{sat} in the laboratory using the falling-head method (ASTM 2010b). Each core was sampled at about 100 mm away from the shrub stems. The distance between the three cores was at least 1.5 m (Fig. 2a). After testing, plant roots were carefully removed from each core to determine R_v using an image-based analysis (Himmelbauer et al. 2004). According to the standardized root washing procedures suggested by Smucker et al. (1982), the soil attached to the roots was washed away with great care following the principle of hydropneumatics elutriation. After placing each soil core into a washing chamber, the soil was dispersed by a hydrovortex that was produced by pressurized jets of water. Soil particles were then separated from roots by flotation in a flow of air bubbles in a horizontal tube. The soil particles remaining in the chamber were then evacuated. This is a much more efficient method than hand separation and could minimize the damage of roots. By measuring root length and root diameter using image analysis (Himmelbauer et al. 2004), total root volume in each core can be indirectly calculated. An average R_v was determined by dividing total root volume over the core volume. η was obtained by dividing the volume of decayed roots by the total root volume. The colour of the decayed roots was dark brown or black, which were distinguishable from the whitish live roots (Fig. 3). Moreover, the decayed roots were much limper and brittle. When pulling a decayed root by hand, the root cortex would be torn off from root endodermis easily (Ni et al. 2017).

Fig. 4. Comparison between measured and fitted or predicted soil water retention curves (SWRCs) from (a) laboratory test and (b) field test. [Colour online.]



In the laboratory test (Ng et al. 2016b), *Schefflera heptaphylla* was used and transplanted to compacted CDG in a steel drum (Fig. 2b). Two plant spacings, 60 mm (S60) and 180 mm (S180), were tested. The soil surface of all drums was exposed to the same, constant atmospheric conditions in the plant room for evaporation and evapotranspiration. After 4 months of growth, all drums were saturated first and then subjected to a 4 day drying period. VWC and suction within the root zone at 50 mm depth were measured by a miniature-tip tensiometers (2100F, Soil Moisture Equipment Corporation) and a soil moisture probe (SM300, Delta-T Device Ltd.), respectively. Then, infiltration tests were carried out to determine k_{sat} . All the drums in the laboratory were subjected to a rainfall event with an intensity of 73 mm/h for 2 h (corresponding to a return period of 10 years in Hong Kong; Lam and Leung 1995) by using a rainfall system described by Ng et al. (2016b). As the amount of rainfall intercepted by leaves was negligible, the infiltration rate was determined by subtracting any surface runoff from the applied rainfall intensity (Ng et al. 2016b). Three replications were tested for each spacing case and for bare soil (as con-

Table 2. Summary of parameters for the newly proposed model.

Test condition	Parameters for eq. (2)					Depth (mm)	A (s/m)	R_v (mm ³ /mm ³)	η (%)
	m_1	m_2	m_3 (kPa)	m_4	e_0				
Laboratory									
Spacing 60 mm	0.15	1.9	0.18	3.51	0.50	100	2.64×10^{-5}	0.044	11
Spacing 180 mm	0.15	1.9	0.18	3.51	0.50	100	2.64×10^{-5}	0.034	1.0
Field									
After 3 months of growth	0.25	1.2	0.7	3.33	0.65	100	3.60×10^{-4}	0.040	1.2
After 10 months of growth	0.25	1.2	0.7	3.33	0.65	100	3.60×10^{-4}	0.053	8.1
After 15 months of growth	0.25	1.2	0.7	3.33	0.65	100	3.60×10^{-4}	0.069	16

trol). The procedures of post-test root excavation and trait characterization were the same as those adopted in the field test.

Statistical analysis was performed using Microsoft Excel. Significant differences were assessed with one way-ANOVA, followed by post hoc Fisher's least significant difference test. Results were considered statistically significant when the p -value ≤ 0.05 .

Observed root growth and decay

After 3 months of plant growth in the field, R_v reached a mean value of $0.04 \text{ mm}^3/\text{mm}^3$ while η was only about 1.2% (Table 1). After 1 year, the roots grew substantially and R_v increased by 72% to $0.07 \text{ mm}^3/\text{mm}^3$ (p -value = 0.07). Simultaneously, more decayed roots were found, causing an increase of η to 16% (p -value < 0.001). More than 80% of the decayed roots were fine roots (i.e., diameter less than 2 mm; Jackson et al. 1997). Fine roots are the main organs for nutrient and water uptake and thus they have high rates of biological activity and hence faster rates of root decay (Raich et al. 2009). Previous studies also found that dead fine roots could be up to one-third of total fine root biomass during plant growth (Espeleta and Clark 2007).

For the laboratory test, after 4 months of growth, R_v for plants spaced at 180 mm was $0.034 \text{ mm}^3/\text{mm}^3$, which was 23% lower than that at 60 mm spacing ($0.044 \text{ mm}^3/\text{mm}^3$; p -value = 0.054; Table 1). The volume of decayed roots increased evidently when plants were closely spaced. η increased from 1% to 11% as plant spacing decreased from 180 to 60 mm (p -value < 0.001). When closely spaced, shrub-shrub competition of water introduced water stress to the plant (Goldberg and Miller 1990). The plants would thus release abscisic acid, a hormone that causes root proliferation (hence increase R_v) for survival (Munns and Sharp 1993).

Model performance

Figure 4a shows the laboratory drying SWRCs. The SWRC of bare soil was best fitted by eq. (2) to calibrate m_1 , m_2 , m_3 , and m_4 (Table 2). Based on these calibrated parameters and known η and R_v (Table 2), SWRCs of vegetated soil, regardless of the plant spacing, can be well captured by eq. (3) ($R^2 > 0.88$), when k in the equation was set to be 10 for the specific soil type and plant family tested. When the plants were widely spaced, less decayed roots were identified and thus the SWRC was more significantly affected by the reduction of void ratio due to root occupancy. For closely spaced plants, more decayed roots were found (11%), consequently resulting in a net increase in void ratio and hence the reduction of water retention capability. Some discrepancies are found for the case of 60 mm spacing, though no more than 13% of S_r in high suction range. This may be because of the empirical restraint of setting the product of $m_1 m_2 m_4$ to be 1 in eq. (2). Due to the lack of data, this empirical relation could not be verified when plant roots exist in soil. Other factors, such as the release of root exudates or organic acid (Grayston et al. 1997; Traoré et al. 2000), may affect the contact angle of water menisci and soil hydrophobicity (Yun et al. 2010; Carminati et al. 2016), and hence the soil water retention and permeability. The predictions made by the proposed equations in this study might be improved if these factors were to be well-understood and taken into account.

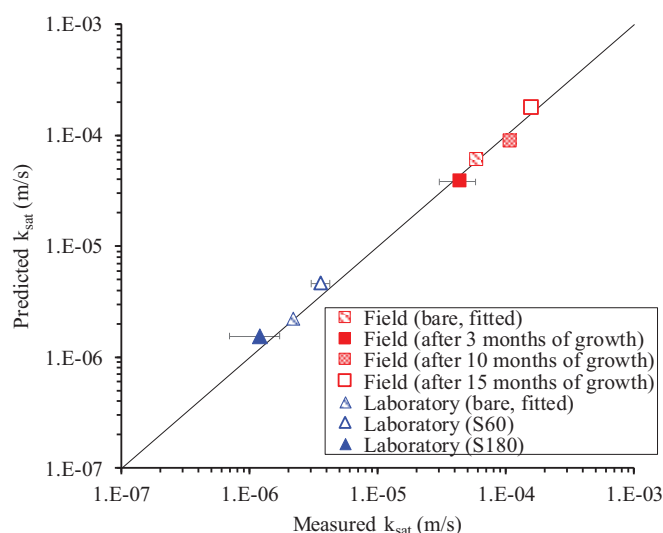
Comparisons of the measured and fitted-predicted field drying SWRCs are shown in Fig. 4b. After calibrating the parameters m_1 , m_2 , m_3 , and m_4 using the SWRC of the bare soil and using the same $k = 10$ for the same soil type and plant family tested (see Table 2), the field SWRCs of vegetated soils at the three growth periods (3, 10, and 15 months) were predicted using the respective values of η and R_v determined in the field (Table 2). Good matches ($R^2 > 0.94$) were found for all three cases. When the plant was relatively young, only a small percentage of roots decayed (1.2%), so in this case, the main mechanism causing the change of SWRC was likely through root occupancy. When the plant became older and had more decayed roots (16%), the increase in soil void ratio outweighed the effects of root occupancy, causing a net reduction of soil water retention capability. Specifically, for the case of 10 month growth, the SWRC of the vegetated soil was quite close to that of the bare case. This is probably because the root occupancy effects may have been almost cancelled out by the effects of root decay.

Figure 5 compares the measured and predicted k_{sat} . The k_{sat} of bare soil, in both field and laboratory tests, was used to calibrate the parameter A in eq. (5) (see Table 2), so the match is expected (Fig. 5). By using the same k of 10, close agreements of measured and predicted k_{sat} of vegetated soils ($R^2 > 0.92$) were obtained in both cases. When compared with the k_{sat} of bare soil, the k_{sat} of vegetated soils could be either lower or higher, depending on the relative weight of the effects of root occupancy and root decay. When the root occupancy is more dominant (e.g., after 3 months of growth for the field test and S180 for the laboratory test), the k_{sat} of vegetated soils is less than that of the bare soil. On the contrary, when the effects of root decay outweigh the root occupancy effects (e.g., after more than 10 months of growth for the field test and S60 for the laboratory test), k_{sat} of vegetated soils would be higher. The test results reveal that the soil water retention and k_{sat} depended on the growth stage of plants. Indeed, it is not uncommon to find a root-induced increase in k_{sat} by 1.3–6.5 times due to root aging effects in the laboratory and the field (Li and Ghodrati 1994; Vergani and Graf 2016; Leung et al. 2018). Previous numerical studies (Shao et al. 2017; Ni et al. 2018) have shown that root-induced increase in k_{sat} could be detrimental to shallow slope stability due to the reduced ability of the vegetated soil to preserve suction during rainfall.

Conclusions

This note proposes new and simple models that can capture the effects of root growth and root decay on both soil water retention curve (SWRC) and saturated permeability (k_{sat}) of coarse-grained soils. A new void ratio function is formulated so that the reduction of soil void ratio (due to root growth) and increase in void ratio (due to root decay) could be modelled. SWRC of vegetated soil can be determined by combining the void ratio function with a void ratio-dependent water retention model. Two specific plant traits, namely root volume ratio and root decay ratio (both can be obtained by careful root excavation and image-based analysis), are needed to quantify the new void ratio function. Moreover, a

Fig. 5. Comparison between measured and predicted k_{sat} of vegetated soils. [Colour online.]



model parameter that is specific to the soil type and plant family tested is also required to describe the rate of change of air-entry value (AEV) due to root growth and root decay. On the contrary, combining the function with the Kozeny–Carman (KC) equation allows the saturated water permeability (k_{sat}) of vegetated soil to be predicted. Based on a set of new field test data and an existing laboratory experiment, good agreements between measurements and prediction are found for the SWRC and k_{sat} of a silty sand vegetated with the plants under the family *Schefflera*.

Acknowledgements

The authors acknowledge the research grant No. 51778166 awarded by the National Natural Science Foundation of China, research grant 2012CB719805 from the National Basic Research Program (973 Program) administered by the Ministry of Science and Technology of the People's Republic of China, and research grants HKUST6/CRF/12R and GRF/16212818 awarded by the Research Grants Council of the Government of the Hong Kong SAR. The second author acknowledges the EU Marie Curie Career Integration Grant under the project “BioEPIC slope”.

References

ASTM. 2010a. Standard practice for classification of soils for engineering purposes (Unified Soil Classification System). ASTM standard D2487. American Society for Testing and Materials, West Conshohocken, Pa. doi:10.1520/D2487-11.

ASTM. 2010b. Standard test methods for measurement of hydraulic conductivity of saturated porous materials using a flexible wall permeameter. ASTM standard D5084. West Conshohocken, Pa. doi:10.1520/D5084-00.

Ball, D.F. 1964. Loss-on-ignition as an estimate of organic matter and organic carbon in non-calcareous soils. *European Journal of Soil Science*, **15**(1): 84–92. doi:10.1111/j.1365-2389.1964.tb00247.x.

Block, R.M.A., Rees, K.C.J., and Knight, J.D. 2006. A review of fine root dynamics in *Populus* plantations. *Agroforestry Systems*, **67**: 73–84. doi:10.1007/s10457-005-2002-7.

Boldrin, D., Leung, A.K., and Bengough, A.G. 2017. Root biomechanical properties during establishment of woody perennials. *Ecological Engineering*, **109**: 196–206. doi:10.1016/j.ecoleng.2017.05.002.

Boldrin, D., Leung, A.K., and Bengough, A.G. 2018. Effects of root dehydration on bio-mechanical properties of woody roots of *U. europaeus*. *Plant and Soil*, **431**: 347–369. doi:10.1007/s11104-018-3766-7.

Brunner, I., Herzog, C., Dawes, M.A., Arend, M., and Sperisen, C. 2015. How tree roots respond to drought. *Frontiers in Plant Science*, **6**(547): 547. PMID: 26284083.

Carman, P.C. 1938. The determination of the specific surface of powders. *Journal of the Society of Chemical Industries, Transactions*, **57**: 225.

Carminati, A., Zarebanadkouki, M., Kroener, E., Ahmed, M.A., and Holz, M. 2016. Biophysical rhizosphere processes affecting root water uptake. *Annals of Botany*, **118**: 561–571. doi:10.1093/aob/mcw113.

Carrier, W.D. 2003. Goodbye, Hazen; Hello, Kozeny–Carman. *Journal of Geotechnical and Geoenvironmental Engineering*, **129**(11): 1054–1056. doi:10.1061/(ASCE)1090-0241(2003)129:11(1054).

Chapuis, R.P., and Aubertin, M. 2003. On the use of the Kozeny–Carman equation to predict the hydraulic conductivity of soils. *Canadian Geotechnical Journal*, **40**(3): 616–628. doi:10.1139/t03-013.

Chen, X.W., Wong, J.T.F., Chen, Z.T., Tang, T.W.L., Guo, H.W., Leung, A., et al. 2018. Effects of biochar on the ecological performance of a subtropical landfill. *Science and the Total Environment*, **644**: 963–975. doi:10.1016/j.scitotenv.2018.06.379.

Delta-T Devices, Ltd. 2010. User manual for the SM300 Soil moisture sensor. Cambridge, U.K.

Eissenstat, D.M., Wells, C.E., Yanai, R.D., and Whitbeck, J.L. 2000. Building roots in a changing environment: implications for root longevity. *New Phytologist*, **147**: 33–42. doi:10.1046/j.1469-8137.2000.00686.x.

Espeleta, J.F., and Clark, D.A. 2007. Multi-scale variation in fine-root biomass in a tropical rain forest: a seven-year study. *Ecological Monographs*, **77**(3): 377–404. doi:10.1890/06-1257.1.

Gallipoli, D., Wheeler, S.J., and Karstunen, M. 2003. Modelling the variation of degree of saturation in a deformable unsaturated soil. *Géotechnique*, **53**(1): 105–112. doi:10.1680/geot.2003.53.1.105.

GEO. 2011. Technical guidelines on landscape treatment for slopes. Geotechnical Engineering Office (GEO), Hong Kong, China.

Ghestem, M., Sidle, R.C., and Stokes, A. 2011. The influence of plant root systems on subsurface flow: implications for slope stability. *Bioscience*, **61**(11): 869–879. doi:10.1525/bio.2011.61.11.6.

Gish, T.J., and Jury, W.A. 1983. Effect of plant roots and root channels on solute transport. *Transactions of the ASAE*, **26**(2): 440–444. doi:10.13031/2013.33955.

Goldberg, D.E., and Miller, T.E. 1990. Effects of different resource additions on species diversity in an annual plant community. *Ecology*, **71**(1): 213–225. doi:10.2307/1940261.

Grayston, S.J., Vaughan, D., and Jones, D. 1997. Rhizosphere carbon flow in trees, in comparison with annual plants: the importance of root exudation and its impact on microbial activity and nutrient availability. *Applied Soil Ecology*, **5**(1): 29–56. doi:10.1016/S0929-1393(96)00126-6.

Himmelbauer, M.L., Loiskandl, W., and Kastanek, F. 2004. Estimating length, average diameter and surface area of roots using two different image analysis systems. *Plant and Soil*, **260**(1): 111–120. doi:10.1023/B:PLSO.0000030171.28821.55.

Jackson, R.B., Mooney, H.A., and Schulze, E.D. 1997. A global budget for fine root biomass, surface area, and nutrient contents. *Proceedings of the National Academy of Sciences*, **94**: 7362–7366. doi:10.1073/pnas.94.14.7362.

Jacobs, L.M., Sulman, B.N., Brzostek, E.R., Feighery, J.J., and Phillips, R.P. 2018. Interactions among decaying leaf litter, root litter and soil organic matter vary with mycorrhizal type. *Journal of Ecology*, **106**(2): 502–513. doi:10.1111/1365-2745.12921.

Jassogne, L., McNeill, A., and Chittleborough, D. 2007. 3D-visualization and analysis of macro- and meso-porosity of the upper horizontal of a sodic, texture-contrast soil. *European Journal of Soil Science*, **58**(3): 589–598. doi:10.1111/j.1365-2389.2006.00849.x.

Jotisankasa, A., and Sirirattanachai, T. 2017. Effects of grass roots on soil-water retention curve and permeability function. *Canadian Geotechnical Journal*, **54**(11): 1612–1622. doi:10.1139/cgj-2016-0281.

Lam, C.C., and Leung, Y.K. 1995. Extreme rainfall statistics and design rainstorm profiles at selected locations in Hong Kong. *Royal Observatory, Hong Kong*.

Leung, A.K., Garg, A., Co, J.L., Ng, C.W.W., and Hau, B.C.H. 2015a. Effects of the roots of *Cynodon dactylon* and *Schefflera heptaphylla* on water infiltration rate and soil hydraulic conductivity. *Hydrological Processes*, **29**(15): 3342–3354. doi:10.1002/hyp.10452.

Leung, A.K., Garg, A., and Ng, C.W.W. 2015b. Effects of plant roots on soil–water retention and induced suction in vegetated soil. *Engineering Geology*, **193**: 183–197. doi:10.1016/j.enggeo.2015.04.017.

Leung, A.K., Boldrin, D., Liang, T., Wu, Z., Kamchoom, V., and Bengough, A.G. (2018). Plant age effects on soil infiltration rate during early plant establishment. *Géotechnique*, **68**(7): 646–652.

Li, Y., and Ghodrati, M. 1994. Preferential transport of nitrate through soil columns containing root channels. *Soil Science Society of America Journal*, **58**: 653–9. doi:10.2136/sssaj1994.03615995005800030003x.

Mitchell, A.R., Ellsworth, T.R., and Meek, B.D. 1995. Effect of root systems on preferential flow in swelling soil. *Communications in Soil Science and Plant Analysis*, **26**(15–16): 2655–2666. doi:10.1080/00103629509369475.

Munns, R., and Sharp, R.E. 1993. Involvement of abscisic acid in controlling plant growth in soil of low water potential. *Functional Plant Biology*, **20**(5): 425–437. doi:10.1071/PP9930425.

Ng, C.W.W., Ni, J.J., Leung, A.K., and Wang, Z.J. 2016a. A new and simple water retention model for root-permeated soils. *Géotechnique Letters*, **6**(1): 106–111. doi:10.1680/jgele.15.00187.

Ng, C.W.W., Ni, J.J., Leung, A.K., Zhou, C., and Wang, Z.J. 2016b. Effects of planting density on tree growth and induced soil suction. *Géotechnique*, **66**(9): 711–724. doi:10.1680/jgeot.15.P.196.

Ni, J.J., Leung, A.K., and Ng, C.W.W. 2017. Investigation of plant growth and transpiration-induced matric suction under mixed grass-tree conditions. *Canadian Geotechnical Journal*, **54**(4): 561–573. doi:10.1139/cgj-2016-0226.

Ni, J.J., Leung, A.K., Ng, C.W.W., and Shao, W. 2018. Modelling hydro-mechanical

- reinforcements of vegetation to slope stability. *Computers and Geotechnics*, **95**: 99–109. doi:10.1016/j.compgeo.2017.09.001.
- Rahardjo, H., Satyanaga, A., Leong, E.C., Santoso, V.A., and Ng, Y.S. 2014. Performance of an instrumented slope covered with shrubs and deep-rooted grass. *Soils and Foundations*, **54**(3): 417–425. doi:10.1016/j.sandf.2014.04.010.
- Raich, J.W., Russell, A.E., and Valverde-Barrantes, O. 2009. Fine root decay rates vary widely among lowland tropical tree species. *Oecologia*, **161**: 325–330. doi:10.1007/s00442-009-1379-9. PMID:19484478.
- Scanlan, C.A., and Hinz, C. 2010. Insight into the processes and effects of root induced changes to soil hydraulic properties. In *Proceedings of the 19th World Congress of Soil Science, Soil Solutions for a Changing World*, Brisbane, Australia. Edited by R. Gilkes and N. Prakongkep, pp. 41–44. [CD-ROM].
- Scholl, P., Leitner, D., Kammerer, G., Loiskandl, W., Kaul, H.P., and Bodner, G. 2014. Root induced changes of effective 1D hydraulic properties in a soil column. *Plant and Soil*, **381**(1–2): 193–213. doi:10.1007/s11104-014-2121-x. PMID:25834290.
- Shao, W., Ni, J.J., Leung, A.K., Su, Y., and Ng, C.W.W. 2017. Analysis of plant root-induced preferential flow and pore-water variation by a dual-permeability model. *Canadian Geotechnical Journal*, **54**(11): 1537–1552. doi:10.1139/cgj-2016-0629.
- Smith, S.W., Woodin, S.J., Pakeman, R.J., Johnson, D., and van der Wal, R. 2014. Root traits predict decomposition across a landscape-scale grazing experiment. *New Phytologist*, **203**(3): 851–862. doi:10.1111/nph.12845. PMID:24841886.
- Smucker, A.J.M., McBurney, S.L., and Srivastava, A.K. 1982. Quantitative separation of roots from compacted soil profiles by the hydropneumatic elutriation system. *Agronomy Journal*, **74**(3): 500–503. doi:10.2134/agronj1982.00021962007400030023x.
- Soilmoisture Equipment Corporation. 1997. Model 2725A jet-filled tensiometer operation instructions. Soilmoisture Equipment Corp., Santa Barbara, Calif., USA.
- Traoré, O., Groleau-Renaud, V., Plantureux, S., Tubeileh, A., and Boeuf-Tremblay, V. 2000. Effect of root mucilage and modelled root exudates on soil structure. *European Journal of Soil Science*, **51**(4): 575–581. doi:10.1111/j.1365-2389.2000.00348.x.
- Urumović, K., and Urumović, K.S. 2016. The referential grain size and effective porosity in the Kozeny-Carman model. *Hydrology and Earth System Sciences*, **20**(5): 1669–1680. doi:10.5194/hess-20-1669-2016.
- van Genuchten, M.T. 1980. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal*, **44**(5): 892–898. doi:10.2136/sssaj1980.03615995004400050002x.
- van Noordwijk, M., Widiyanto, H.M., and Hairah, K. 1991. Old tree root channels in acid soils in the humid tropics: Important for crop root penetration, water infiltration and nitrogen management. *Plant and Soil*, **134**(1): 37–44. doi:10.1007/BF00010715.
- Vergani, C., and Graf, F. 2016. Soil permeability, aggregate stability and root growth: a pot experiment from a soil bioengineering perspective. *Ecohydrology*, **9**(5): 830–842. doi:10.1002/eco.1686.
- Whalley, W.R. 1993. Considerations on the use of time-domain reflectometry (TDR) for measuring soil moisture content. *Journal of Soil Science*, **44**: 1–9. doi:10.1111/j.1365-2389.1993.tb00429.x.
- Yan, W.M., and Zhang, G.H. 2015. Soil–water characteristics of compacted sandy and cemented soils with and without vegetation. *Canadian Geotechnical Journal*, **52**(9): 1331–1334. doi:10.1139/cgj-2014-0334.
- Yun, Z., Min, G., Kai, S., Zhou, Y.H., and Yu, J.Q. 2010. Effects of aqueous root extracts and hydrophobic root exudates of cucumber (*Cucumis sativus* L.) on nuclei DNA content and expression of cell cycle-related genes in cucumber radicles. *Plant & Soil*, **327**(1–2): 455–463. doi:10.1007/s11104-009-0075-1.

Appendix A

Fig. A1. Calibration curves of soil moisture probe with different soil organic contents. CDG, completely decomposed granite. [Colour online.]

